MLOps Assignment 01: End-to-End Heart Disease Risk Prediction System

GitHub URL: `https://github.com/arun5991/heart-disease-mlops`

1. Executive Summary & Architecture This project builds and deploys a machine learning API to predict heart disease. The pipeline includes data cleaning, feature engineering, model tuning, MLflow tracking, automated CI/CD via GitHub Actions, Docker containerization, and Kubernetes deployment with Prometheus monitoring.

2. Environment Setup The project uses a Python virtual environment with pinned dependencies in `requirements.txt` to ensure reproducibility.

```
(base) s2025ct05013@ip-10-3-17-165:~/workspace/Assignment1/heart-disease-mlops$ python3 -m venv venv
source venv/bin/activate   # On Windows: venv\Scripts\activate
pip install --upgrade pip
pip install -r requirements.txt
Requirement already satisfied: pip in ./venv/lib/python3.12/site-packages (24.3.1)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 74.7 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.3.1
    Uninstalling pip-24.3.1:
      Successfully uninstalled pip-24.3.1
Successfully installed pip-26.2.1
Collecting numpy==1.26.4 (from -r requirements.txt (line 1))
  Downloading numpy-1.26.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
Collecting pandas==2.2.2 (from -r requirements.txt (line 2))
  Downloading pandas-2.2.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (19 kB)
Collecting scikit-learn==1.5.1 (from -r requirements.txt (line 3))
  Downloading scikit_learn-1.5.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (12 kB)
Collecting matplotlib==3.9.0 (from -r requirements.txt (line 4))
  Downloading matplotlib-3.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (11 kB)
Collecting seaborn==0.13.2 (from -r requirements.txt (line 5))
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Collecting mlflow==2.14.3 (from -r requirements.txt (line 6))
  Downloading mlflow-2.14.3-py3-none-any.whl.metadata (29 kB)
Collecting fastapi==0.111.0 (from -r requirements.txt (line 7))
  Downloading fastapi-0.111.0-py3-none-any.whl.metadata (25 kB)
Collecting uvicorn==0.30.1 (from -r requirements.txt (line 8))
  Downloading uvicorn-0.30.1-py3-none-any.whl.metadata (6.3 kB)
Collecting pydantic==2.8.2 (from -r requirements.txt (line 9))
  Downloading pydantic-2.8.2-py3-none-any.whl.metadata (125 kB)
Collecting prometheus-client==0.20.0 (from -r requirements.txt (line 10))
  Downloading prometheus_client-0.20.0-py3-none-any.whl.metadata (1.8 kB)
Collecting httpx==0.27.0 (from -r requirements.txt (line 11))
  Downloading httpx-0.27.0-py3-none-any.whl.metadata (7.2 kB)
Collecting requests==2.32.3 (from -r requirements.txt (line 12))
  Downloading requests-2.32.3-py3-none-any.whl.metadata (4.6 kB)
Collecting joblib==1.4.2 (from -r requirements.txt (line 13))
  Downloading joblib-1.4.2-py3-none-any.whl.metadata (5.4 kB)
Collecting pytest==8.2.2 (from -r requirements.txt (line 14))
  Downloading pytest-8.2.2-py3-none-any.whl.metadata (7.6 kB)
Collecting flake8==7.1.0 (from -r requirements.txt (line 15))
  Downloading flake8-7.1.0-py2.py3-none-any.whl.metadata (3.8 kB)
Collecting python-dateutil>=2.8.2 (from pandas==2.2.2->-r requirements.txt (line 2))
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas==2.2.2->-r requirements.txt (line 2))
  Downloading pytz-2026.5-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas==2.2.2->-r requirements.txt (line 2))
  Downloading tzdata-2026.5-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting scipy>=1.6.0 (from scikit-learn==1.5.1->-r requirements.txt (line 3))
  Downloading scipy-1.18.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
Collecting threadpoolctl>=3.1.0 (from scikit-learn==1.5.1->-r requirements.txt (line 3))
  Downloading threadpoolctl-3.7.0-py3-none-any.whl.metadata (24 kB)
```

3. Data Acquisition & Exploratory Data Analysis (EDA) The UCI Heart Disease dataset was cleaned by imputing missing values and binarizing the target variable. The dataset is well-balanced.

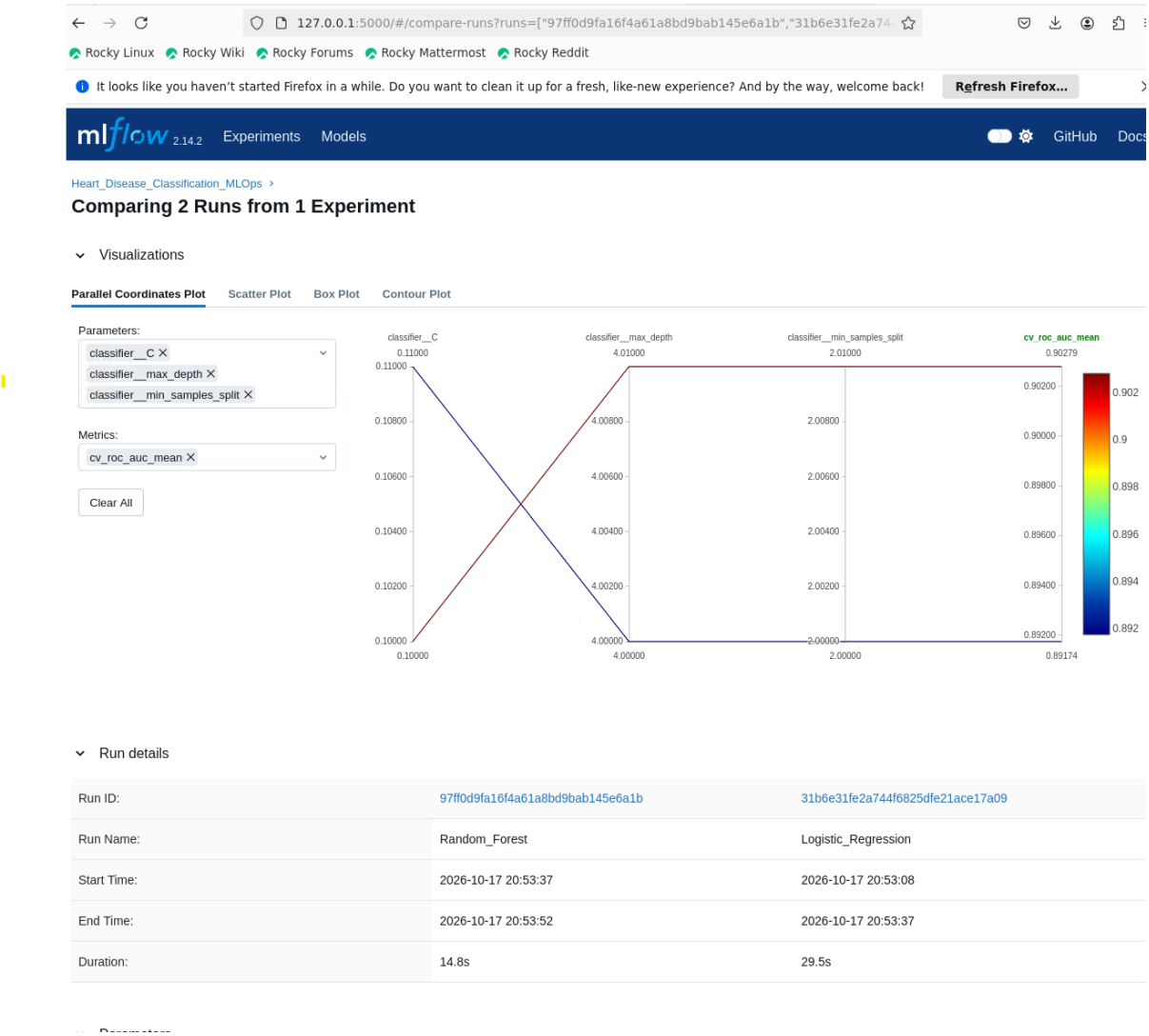
```
(venv) (base) s2025ct05013@ip-10-3-17-165:~/workspace/Assignment1/heart-disease-mlops$ python notebooks_and_scri
pts/01_data_acquisition_and_eda.py
Downloading UCI Heart Disease dataset from https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disea
se/processed.cleveland.data...
Saved raw dataset to data/raw/heart_uci_raw.csv
=== Dataset Shape & Info ===
Total rows: 303, Total columns: 14

Missing values after cleaning:
 age         0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
target       0
dtype: int64

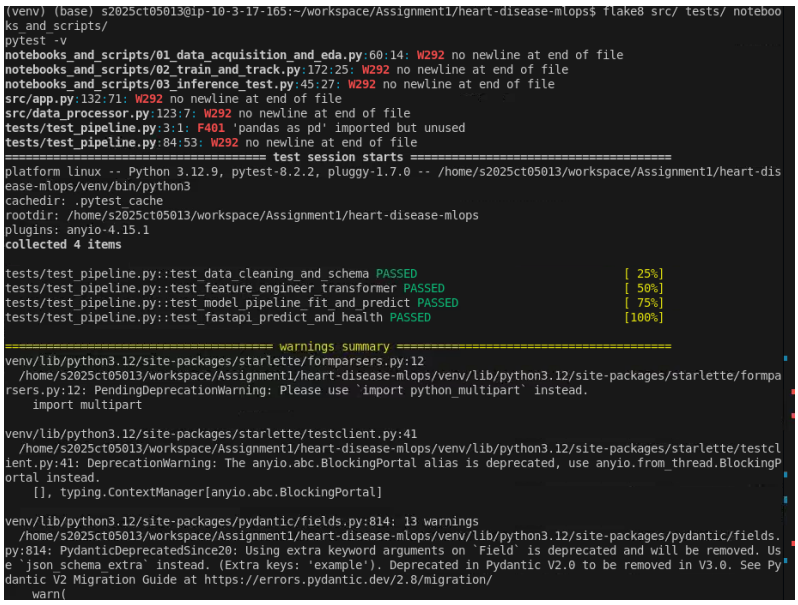
Target Class Distribution:
 target
0    164
1    139
Name: count, dtype: int64

EDA plots saved successfully to 'artifacts/plots/'
```

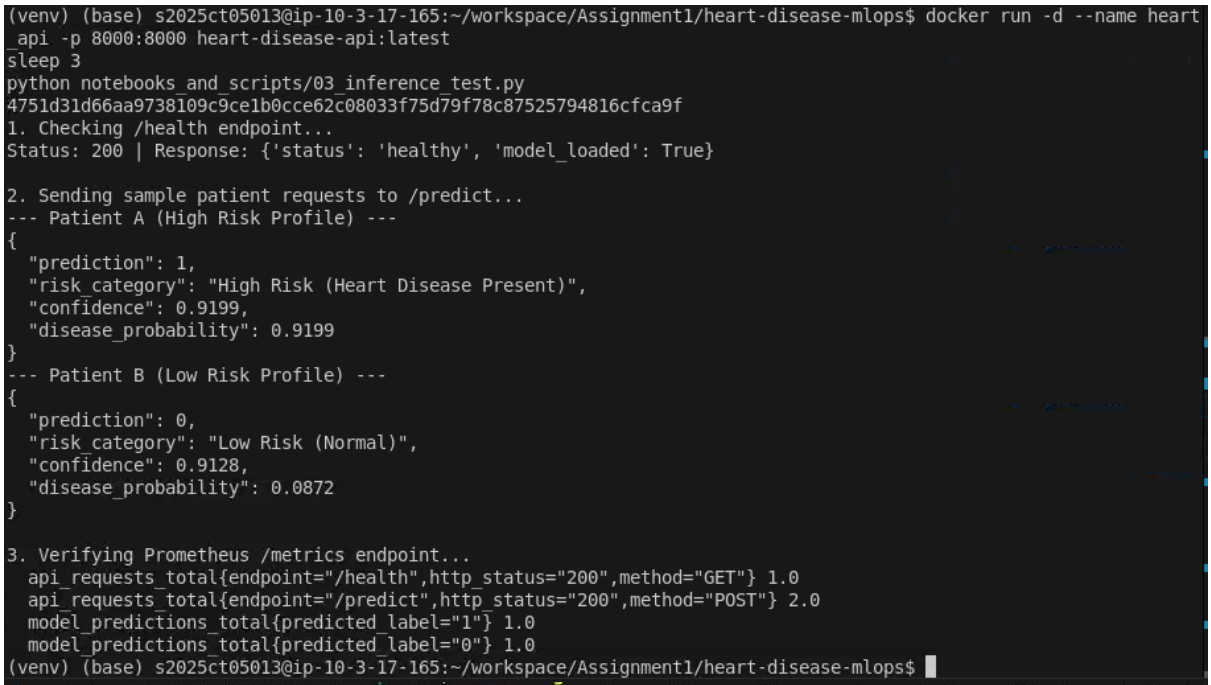
4. Model Development & MLflow Tracking Logistic Regression and Random Forest models were trained and tuned. All parameters, metrics (Accuracy, ROC-AUC), and plots were tracked using MLflow.

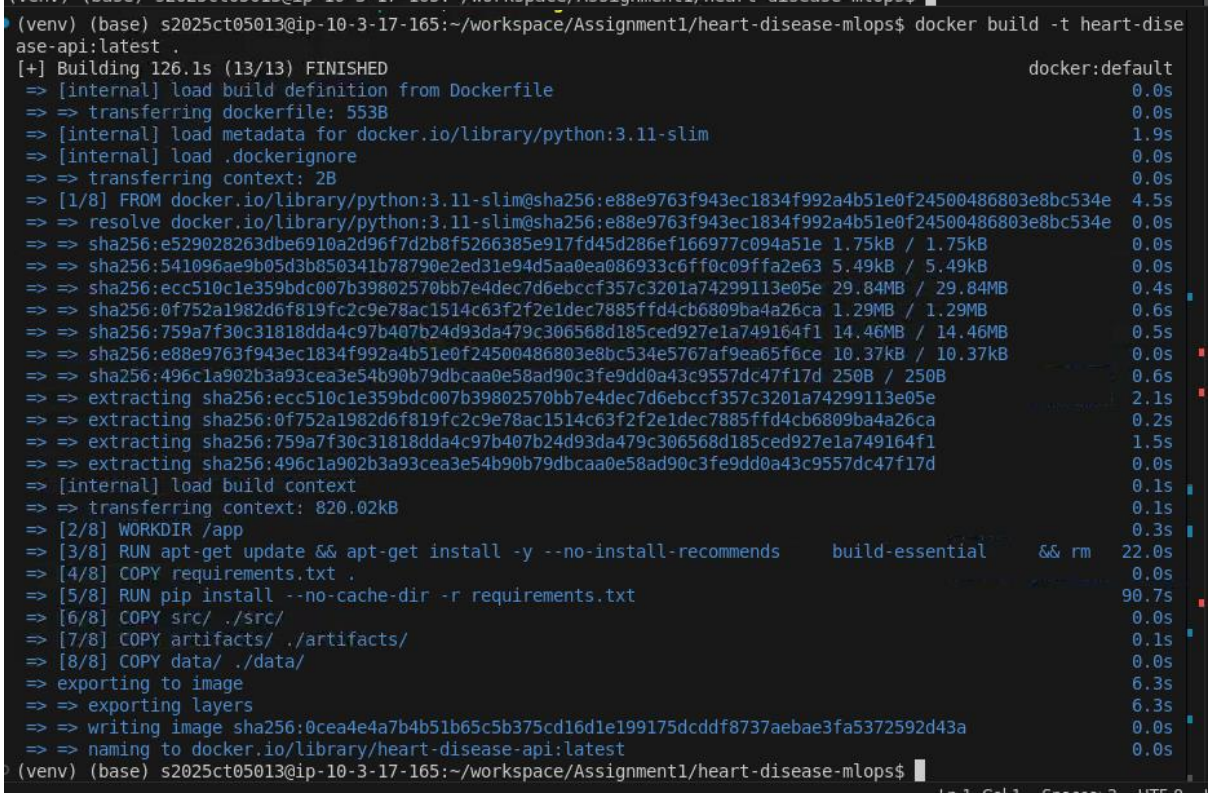


5. Automated Testing & CI/CD The code was linted with Flake8 and tested with Pytest. A GitHub Actions workflow automates testing and container building on every push.

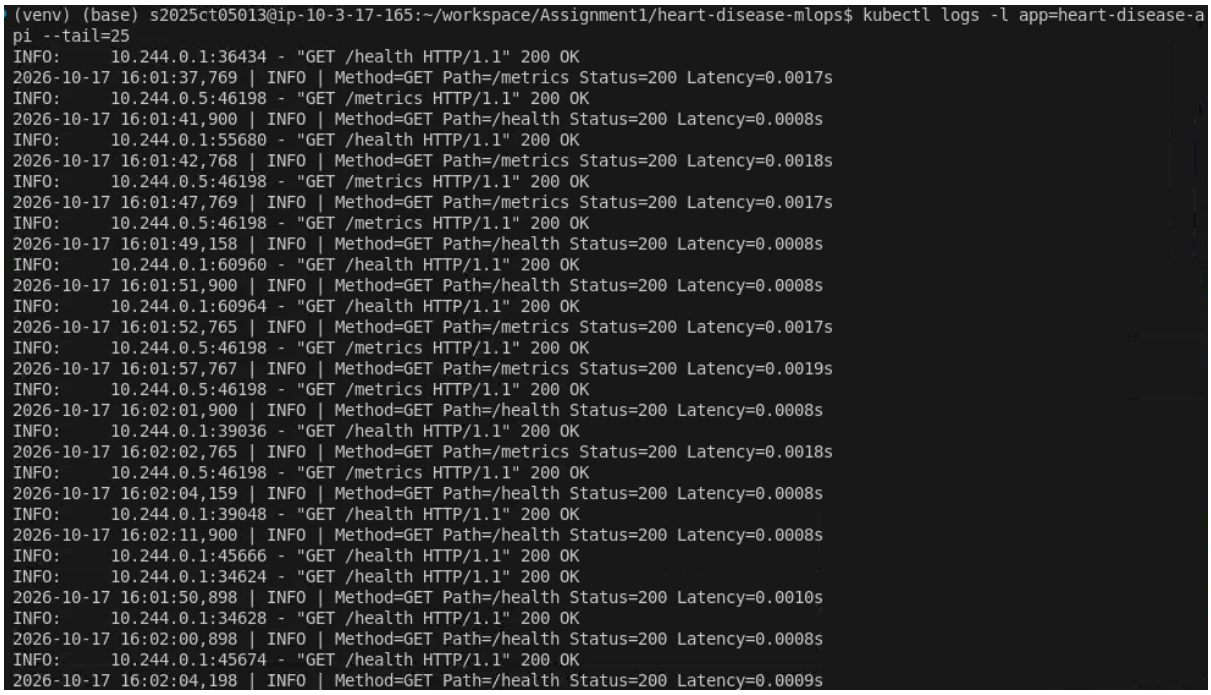


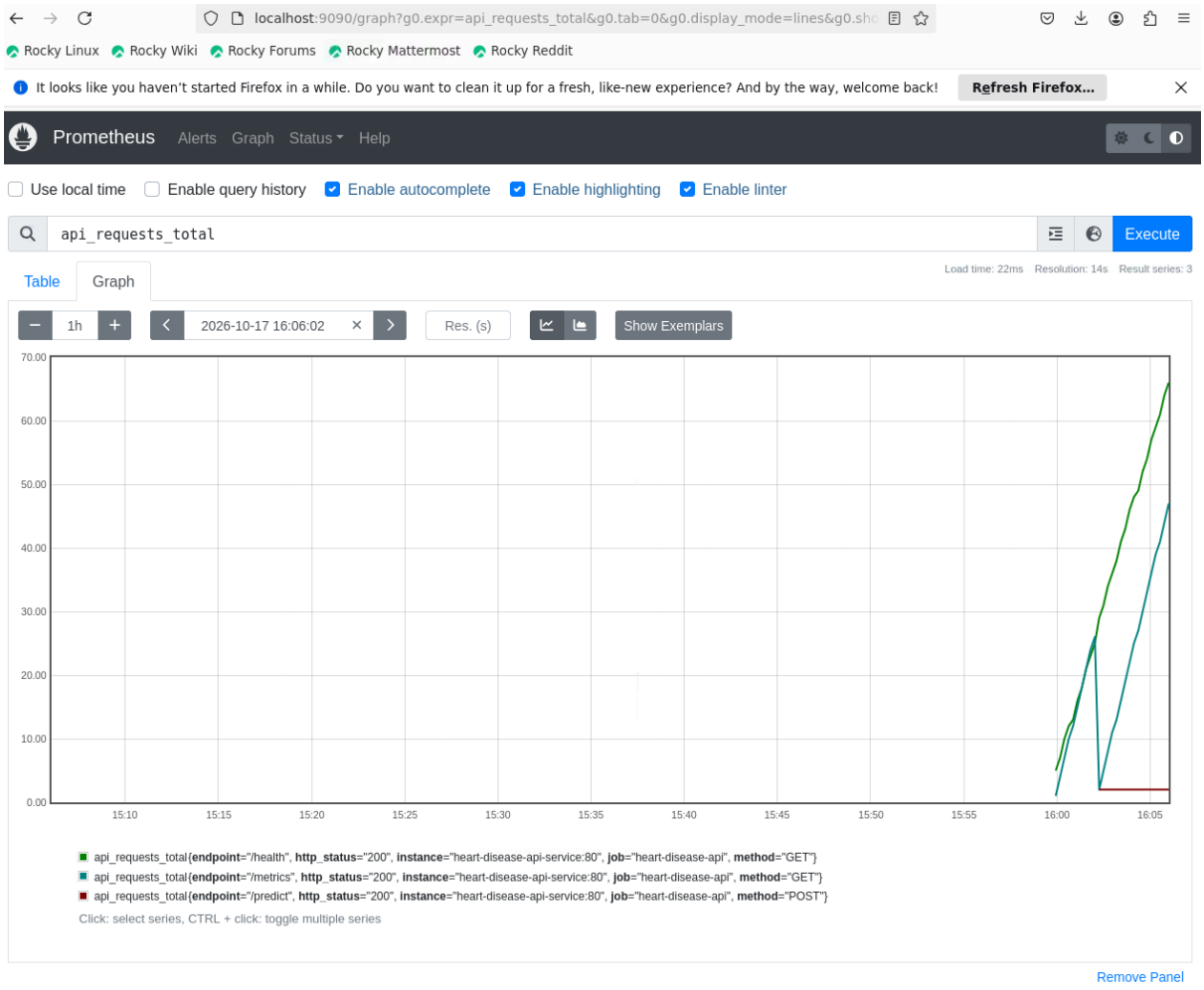
6. Docker Containerization The model is served via a FastAPI application packaged inside a Docker container.





7. Kubernetes Deployment & Monitoring The container is deployed to Kubernetes using a LoadBalancer. API requests are logged, and Prometheus scrapes the `/metrics` endpoint to monitor traffic.






```
(venv) (base) s2025ct05013@ip-10-3-17-165:~/workspace/Assignment1/heart-disease-mlops$ kubectl get deployments,po
ds,svc,ingress
NAME                                           READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/heart-disease-api-deployment   2/2     2            2           61s
deployment.apps/prometheus-deployment          1/1     1            1           47s

NAME                                                READY   STATUS    RESTARTS   AGE
pod/heart-disease-api-deployment-5bcb5f5797-67pvl   1/1     Running   0          61s
pod/heart-disease-api-deployment-5bcb5f5797-f5xpb   1/1     Running   0          61s
pod/prometheus-deployment-5c9f556dc8-x8pgk          1/1     Running   0          47s

NAME                                TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/heart-disease-api-service   LoadBalancer   10.110.143.239   <pending>     80:30080/TCP     61s
service/kubernetes                  ClusterIP      10.96.0.1        <none>        443/TCP          3m14s
service/prometheus-service          NodePort       10.101.224.113   <none>        9090:30090/TCP   47s

NAME                                                    CLASS    HOSTS                ADDRESS   PORTS   AGE
ingress.networking.k8s.io/heart-disease-api-ingress   <none>   heart-mlops.local              80      61s
(venv) (base) s2025ct05013@ip-10-3-17-165:~/workspace/Assignment1/heart-disease-mlops$
```